Numerical Computing Project

Prof. Dr. Muhammad Arif Hussain

Goal: We know that learning is enhanced when students consider problems from different perspectives. So, goal is to delve deeply into a topic of interest by finding and studying an article or part of a book on that topic and then writing a report, which should include some mathematical analysis and numerical computations.

The project report and presentation is your opportunity to learn about something that involves some aspect of numerical computing. Choose a project title we discussed in the class.

Timeline:

• Choose a topic by April 18 and email me a proposal of what you want to do (a few sentences describing your proposed project).

• Almost all projects should include programs in Matlab and Python for the solution of problems.

• Final report (hard copy) due May 10. Emailing me your soft copy of completed work.

Report guidelines:

The report should be roughly 10-12 pages double-spaced, using Word format. The report should include significant mathematics (theoretical or computational). It should also include at least three real world applications. MATLAB and PYTHON codes may be included as part of the report.

Sources:

You should use at least two sources of information, which may include your textbook, other books, and scholarly articles. You should not rely on a website as a main source of information (since websites often contain incorrect information), but searching the web may be helpful initially as an idea-generator of interesting topics and for basic information.

Your report should list all sources used in to writing your report. You may use any standard style to cite them, for example:

Baker, G.L., and Gollub, J.P. Mathematical Modeling: An Introduction, Cambridge University Press, Cambridge, 2020.

Li, T.-y., and Yorke, J., “Iterative methods.” American Mathematical Monthly 82 (1999), 985-992.

If you copy a figure, cite the source in the caption.